

Patient-Reported Outcomes

Understanding the Transition Between Age-Specific Measures of Health-Related Quality of Life: Evidence on the Relationship Between and Comparative Performance of the EQ-5D-Y-5L and EQ-5D-5L

Nicole Reyes, MPH, Tianxin Pan, PhD, Renee Jones, MPH, Kim Dalziel, PhD, Nancy Devlin, PhD, on behalf of the Quality of Life in Kids: Key evidence to strengthen decisions in Australia (QUOKKA) project team

ABSTRACT

Objectives: The EQ-5D-5L is widely used to measure adults' health-related quality of life (HRQoL). The EQ-5D-Y-5L is a corresponding measure adapted for children/adolescents, in principle allowing HRQoL to be measured consistently from childhood to adulthood. However, little is known about how their measurement properties compare. This study investigated the relationship between EQ-5D-Y-5L and EQ-5D-5L in adolescents and compared their psychometric performance.

Methods: The Australian Pediatric Multi-Instrument Comparison Study includes a sample of 591 adolescents (aged 12-18) who completed both EQ-5D-5L and EQ-5D-Y-5L. Responses were compared descriptively and HRQoL summarized using the level sum score. Acceptability, feasibility, ceiling effects, convergence, test-retest reliability, and known-group validity were assessed overall and in subgroups defined by special healthcare needs (SHCN), mental health concerns (MHC), and age (12-13, 14-16, and 17-18 years).

Results: Ceiling effects were lower for EQ-5D-Y-5L than EQ-5D-5L. The EQ-5D-Y-5L better differentiated between adolescents with and without SHCN and MHC than EQ-5D-5L, whereas EQ-5D-5L showed better test-retest reliability in adolescents with SHCN and MHC. We found strong correlations between dimensions anticipated to be correlated. EQ-5D-Y-5L identified a higher incidence of self-reported HRQoL problems than EQ-5D-5L both overall and particularly in mental health.

Conclusions: Although both instruments are valid for measuring HRQoL in adolescents aged 12 to 18 years, EQ-5D-Y-5L had some psychometric advantages. The instruments are closely related, but differences in their descriptive systems produce differences in self-reported HRQoL. Results highlight potential discontinuities in HRQoL measured using age-specific instruments, which may be important for their use in economic models that involve transitions between age groups.

Keywords: Australia, EQ-5D-5L, EQ-5D-Y-5L, health-related quality of life, psychometric performance, Adolescents, Pediatrics, Child health.

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Highlights

- Age-specific health-related quality of life (HRQoL) instruments describe health in ways that are appropriate to a given age group. However, measuring health in an age-specific manner means that economic evaluation models that involve transitions between childhood and adulthood may need to address differences in how HRQoL is measured. There is little evidence to inform this.
- We show systematic differences between the way 2 conceptually related instruments, EQ-5D-Y-5L for children and EQ-5D-5L for adults, measure HRQoL in an adolescent population in which both can be used. These differences may be related to the way health problems are described, eg, "worried, sad, or unhappy" versus "anxious/depressed."
- The greater availability of pediatric measures of HRQoL has improved researchers' ability to measure child HRQoL in valid ways. However, applications that involve measuring and comparing HRQoL across child and adult populations cannot assume measurement comparability between age-specific instruments.

Introduction

Health-related quality of life (HRQoL) plays an important role in the economic evaluation of health interventions and health technology assessment worldwide.¹ Methods for measuring and valuing HRQoL for health technology assessment in adults are widely accepted.² More recently, research has also focused on the measurement and valuation of HRQoL in children and adolescents. A number of child-specific generic HRQoL instruments are available.^{3,4} Although the comparative performance of these instruments has been studied,⁵⁻⁷ there remain challenges about how best to measure child and adolescent HRQoL and how to use these instruments in economic evaluation.

One question concerns how to measure HRQoL in cost-effectiveness analyses of pediatric technologies that involve modeling HRQoL from childhood into adulthood.

Such models require evidence on HRQoL generated from different instruments developed for children and adults, respectively, and rely on the assumption that the resulting measurement of HRQoL is broadly comparable. EuroQol instruments have a potential advantage in this respect. The EQ-5D-5L (5L) is a widely used measure of adult HRQoL worldwide,⁸ comprising 5 dimensions and 5 levels. The EQ-5D-Y-5L (Y-5L) is an instrument adapted to be

valid for use in children and adolescents⁹; it has 5 levels, expanding the 3 levels in the EQ-5D-Y-3L (Y-3L). The availability of child and adult HRQoL instruments with the same 5-dimension and 5-level structure, in principle, allows consistency in the measurement of HRQoL across age groups. However, the adaptations required to make the Y-5L child friendly involved changes to the wording of both the dimensions and level descriptors (Fig. 1). This raises the question of whether these differences in wording produce different results, and how this affects the comparative psychometric performance of each instrument?

One way of understanding this issue is to test both instruments in the age group in which both may be used. The user guide for the Y-5L suggests that both the youth and adult EQ-5D instruments can be used in ages 12 to 15 years, and the adult 5L instrument should be used in age 16 years and above—with the exception of studies that only involve children up to 18 years, in which case the youth instrument can be used.¹⁰ Thus, both instruments are candidates for measuring HRQoL in ages 12 to 18 years. However, there is limited evidence on the comparative performance of these instruments in this age range, a lack of clarity about when measurement of HRQoL should transition from the Y-5L to the 5L, and what the consequences of the transition might be for measurement of HRQoL.

This links to a wider issue concerning the use and application of age-specific HRQoL instruments in economic evaluation and the

comparability of estimates of quality-adjusted life years (QALYs) generated from them. Age-specific instruments differ in terms of what aspects of HRQoL are measured, how data are collected (proxy vs self-complete), and how HRQoL states are valued.¹¹ The complex and multiple transitions involved are illustrated in Figure 2. Each of these can in principle affect QALY estimates: the assessment of HRQoL is a product both of how it is measured and how it is valued. Although numerous local value sets for the 5L exist,¹² value sets for the recently launched Y-5L are planned but not yet available. In the meantime, understanding the differences in measurement properties between the Y-5L and 5L is an important first step to understanding the overall implications for QALY estimation.

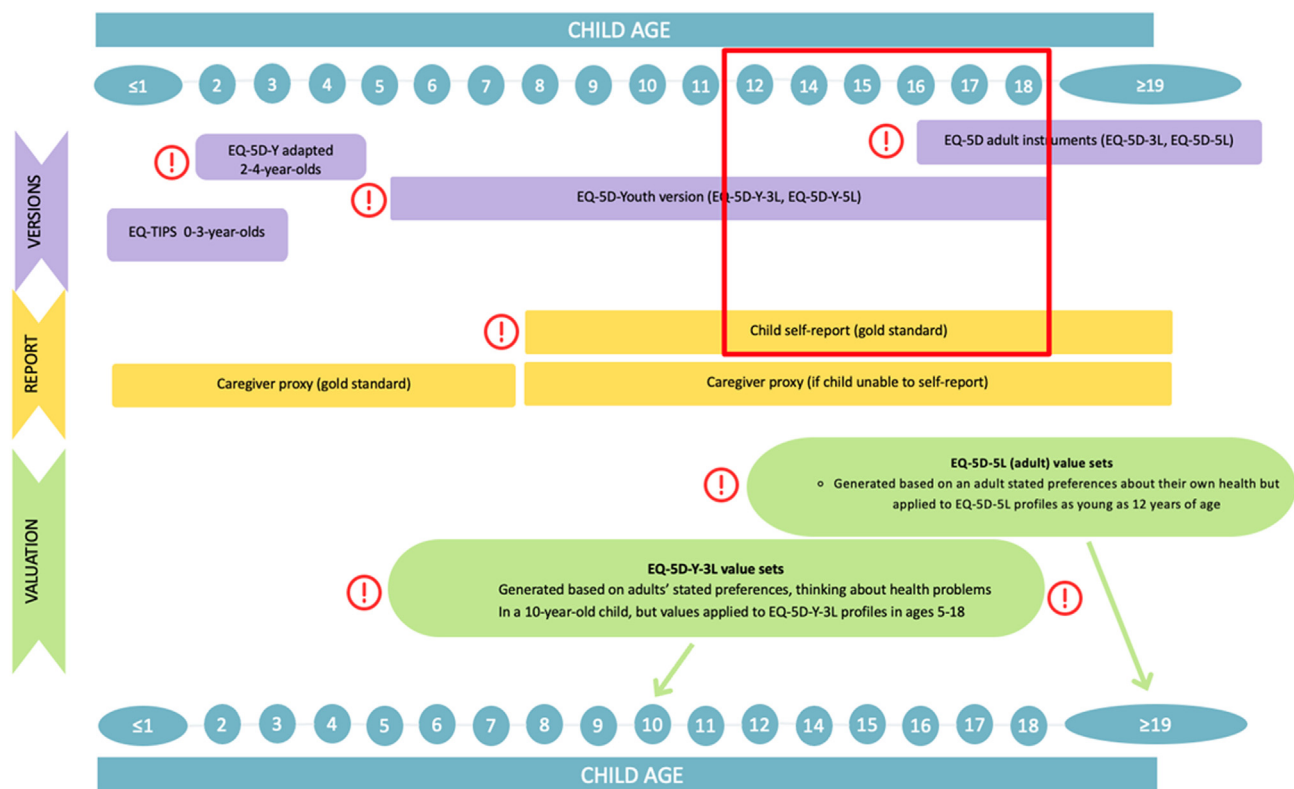
The purpose of this study is to provide evidence on the measurement properties of the Y-5L and the 5L—specifically, the area indicated by the red box in Figure 2. We focused on the 5-level version of these instruments because both the 5L and Y-5L and have shown superior performance compared with the corresponding 3-level versions.^{13,14} The Y-5L is a relatively new instrument (launched September 2024), and evidence on its performance is still developing. This study contributes to the evidence base of the psychometric performance of Y-5L.

The aim of this article is to (1) investigate the relationship between Y-5L and 5L as measures of HRQoL in an age group in which both may be used and (2) compare the psychometric

Figure 1. Sample questionnaire of the EQ-5D-Y-5L (left) and EQ-5D-5L (right).

EQ-5D-Y-5L		EQ-5D-5L	
MOBILITY (walking around)		MOBILITY	
I have no problems in walking around	☐	I have no problems in walking about	☐
I have a little bit of problems walking around	☐	I have slight problems in walking about	☐
I have some problems walking around	☐	I have moderate problems in walking about	☐
I have a lot of problems walking around	☐	I have severe problems in walking about	☐
I cannot walk around	☐	I am unable to walk about	☐
LOOKING AFTER MYSELF		SELF-CARE	
I have no problems washing or dressing myself	☐	I have no problems washing or dressing myself	☐
I have a little bit of problems washing or dressing myself	☐	I have slight problems washing or dressing myself	☐
I have some problems washing or dressing myself	☐	I have moderate problems washing or dressing myself	☐
I have a lot of problems washing or dressing myself	☐	I have severe problems washing or dressing myself	☐
I cannot wash or dress myself	☐	I am unable to wash or dress myself	☐
DOING USUAL ACTIVITIES (for example, going to school, hobbies, sports, playing, doing things with family or friends)		USUAL ACTIVITIES (e.g. work, study, housework, family or leisure activities)	
I have no problems doing my usual activities	☐	I have no problems doing my usual activities	☐
I have little bit of problems doing my usual activities	☐	I have slight problems doing my usual activities	☐
I have some problems doing my usual activities	☐	I have moderate problems doing my usual activities	☐
I have a lot of problems doing my usual activities	☐	I have severe problems doing my usual activities	☐
I cannot do my usual activities	☐	I am unable to do my usual activities	☐
HAVING PAIN OR DISCOMFORT		PAIN / DISCOMFORT	
I have no pain or discomfort	☐	I have no pain or discomfort	☐
I have a little bit of pain or discomfort	☐	I have slight pain or discomfort	☐
I have some pain or discomfort	☐	I have moderate pain or discomfort	☐
I have a lot of pain or discomfort	☐	I have severe pain or discomfort	☐
I have extreme pain or discomfort	☐	I have extreme pain or discomfort	☐
FEELING WORRIED, SAD OR UNHAPPY		ANXIETY / DEPRESSION	
I am not worried, sad or unhappy	☐	I am not anxious or depressed	☐
I am a little bit worried, sad or unhappy	☐	I am slightly anxious or depressed	☐
I am quite worried, sad or unhappy	☐	I am moderately anxious or depressed	☐
I am really worried, sad or unhappy	☐	I am severely anxious or depressed	☐
I am extremely worried, sad or unhappy	☐	I am extremely anxious or depressed	☐

Figure 2. Identifying key transition points between age-specific EQ-5D instruments in terms of instrument version, self-report/proxy, and valuation.



performance of the Y-5L and 5L in a sample of healthy adolescents and in adolescents with special healthcare needs (SHCN) and mental health concerns (MHC).

Methods

Participants

Data for this study were available from the Australian Pediatric Multi-Instrument Comparison (P-MIC) Study, data cut 3 (dated June 27, 2023).¹⁵ Data from adolescents aged 12 to 18 years who completed both the Y-5L and 5L via self-report were included in this analysis. The survey was first provided to caregivers to consent and complete a range of demographic questions. The survey was then handed to the adolescent to consent and complete the HRQoL instruments if the caregiver deemed that they were able to answer questions about their health. The data were collected using an online survey with follow-up at either 2 days or 4 weeks.

Measures and Scoring

The 5L is an adult measure of HRQoL that comprises a descriptive system and a visual analog scale. It measures 5 dimensions: mobility (MO), self-care (SC), usual activities (UA), pain/discomfort (PD), and anxiety/depression (AD). Each dimension has 5 levels: no, slight, moderate, severe, and extreme problems. Self-reported health can be represented by a 5-digit numeric string, eg, no problems across all dimensions is state 11111, extreme problems across all dimensions is state 55555.

The Y-5L is an expanded-level version of the EQ-5D-Y-3L (3L) with 5 rather than 3 levels of problems on each dimension.⁹ The Y-3L was adapted from the adult EQ-5D-3L, covering the similar 5

dimensions and levels but using child-friendly wording.^{16,17} Notable differences in wording between the Y-5L and 5L (Fig. 1) include “worried, sad, or unhappy” (WSU) compared with AD for the 5L. Furthermore, the Y-5L uses “a little bit,” “some/quite,” and “a lot/really” to describe levels 2, 3, and 4 compared with “slight,” “moderate,” and “severe” for the corresponding levels of the 5L.

Both the Y-5L and 5L describe (5^5) 3125 states. We use level sum score (LSS) to summarize health states from both, because there are currently no Y-5L value sets. The LSS provides means of obtaining a score where no value sets are available.¹⁸ It is derived by adding together the digits of the 5-digit health state. For example, health-state 12121 has a LSS of 7. The LSS of both the Y-5L and 5L ranges from 5 to 25.

Groups of Adolescents Used in Analysis

Special healthcare needs

Adolescents with SHCN were defined as those “at increased risk for a chronic physical, developmental, behavioral, or emotional condition and who also require health and related services of a type or amount beyond that required by children generally.”¹⁹ Adolescents were identified using the Children with SHCN screener,²⁰ which has been found to be efficient and reliable.²¹

Mental health concerns

The Strengths and Difficulties Questionnaire (SDQ),²² a 25-item screening questionnaire validated for use in children up to age 17 years, was included in the P-MIC study. Australian SDQ norms were used to identify adolescents with clinically important MHC.²³ For pragmatic reasons, we extended use of SDQ to include individuals aged 18 years. Because the SDQ norms are specific to

only male and female genders, adolescents not identifying as male or female were not included in the MHC subgroup.

The SHCN and MHC groups are not mutually exclusive.

Age groups

The sample was divided into 3 subgroups: 12 to 13, 14 to 16, and 17 to 18 years of age. Analyses were run to determine differences among these age groups across both instruments.

Psychometric Analysis

All analyses were conducted in accordance with the Psychometric Statistical Analysis Plan provided in the P-MIC Technical Methods¹⁵ and conducted using Stata Version 17 (StataCorp). For statistical analysis, a *P* value of $< .05$ was considered statistically significant.

Acceptability and feasibility

Acceptability was assessed using participants' self-reported difficulty completing the measure, on a 5-point scale from 1 being "very difficult" to 5 being "very easy." Differences in difficulty across instruments were assessed using Pearson's chi-square test. All respondents were required to complete all items; there were no missing data.

Feasibility was assessed using the time to complete each instrument reported in seconds. This was recorded by the survey administration platform and was reported descriptively. Time was capped at 10 minutes for each instrument to allow for extreme outliers, which may have been due to people switching to other activities between filling out questionnaires.

Response distribution and ceiling and floor effects

We reported responses distributions in each dimension of the Y-5L and 5L. The relationship between problems reported on the 2 instruments was illustrated using Sankey diagrams. For each instrument, we report the mean and standard deviation (SD), median, and interquartile range (IQR) of the LSS.

Floor and ceiling effects were examined to determine whether an excessive number of responses are clustered at the best or worst health state on each instrument. A ceiling effect is observed if $>15\%$ of respondents with SHCN or with MHCs reported "1111," and a floor effect is observed if $>15\%$ of healthy respondents reported "5555."²⁴

Convergence

Convergence refers to how closely a test is related to another test that measures the same construct. Spearman's correlations were used because data were not normally distributed. Strong correlations (≥ 0.5) were expected between the corresponding dimensions and LSS of the 2 instruments.

Test-retest reliability

Test-retest reliability concerns the consistency of responses among adolescents who reported no change in health comparing their initial and follow-up surveys. In the P-MIC study, the majority of respondents were asked to complete the instruments 4 weeks after the initial observation to allow for a change in health to occur. A small group of respondents were asked to complete the instruments after 2 days to assess test-retest reliability. Because only 14 participants in our sample completed the instruments after 2 days, the test-retest analysis pooled all respondents who completed a follow-up survey either at 2 days or 4 weeks and reported no change in health. At the dimension level, a linear weighted Kappa was used to compare agreement between instruments because the data were ordinal. The weighted kappa

statistics of <0.2 , 0.21–0.40, 0.41–0.60, 0.61–0.8, and >0.80 were considered as poor, fair, moderate, substantial, and almost perfect agreement, respectively.²⁵ At the instrument level, comparison was made via intraclass correlation coefficient (ICC) and the ICC 95% confidence interval (CI). This method is preferred for numerical outcomes and was derived via a 2-way mixed-effects model for a single instrument, based on absolute agreement. ICC values of <0.4 , 0.4–0.59, 0.60–0.74, and ≥ 0.75 were indicative of poor, fair, good, and excellent reliability, respectively.²⁶

Known-group validity

Known-group validity is used to confirm whether the instruments can differentiate between groups that are expected to have differences in HRQoL. Adolescents with SHCN or MHC were identified as groups expected to have lower HRQoL compared with adolescents without SHCN or MHC. We also performed known-group analysis in each age subgroup. Because of the skewed distribution of LSS, we tested the mean difference in LSS between groups using nonparametric Mann-Whitney U test. Effect sizes were estimated using Cohen's *D* with effect sizes 0.2 to 0.49 considered small, 0.5 to 0.79 moderate, and ≥ 0.8 large.²⁷

Summary of psychometric performance

Following the P-MIC Technical Report methods,¹⁵ we summarized the psychometric performance of both instruments. Significant evidence of good performance (indicated by a green tick) for each was based on the following thresholds:

- Ceiling effects: $>15\%$ of participants self-reported 11111.
- Floor effects: $>15\%$ of participants self-reported 55555.
- Convergent validity: strong correlations between dimensions hypothesized to be related (Spearman's correlation ≥ 0.5).
- Test-retest reliability: good or excellent agreement (ICC ≥ 0.6).
- Known-group validity: mean difference with a *P* value of $< .05$ and Cohen's *d* effect size ≥ 0.8 .

Results

Data from 591 adolescents aged 12 to 18 were included in this study. Appendix Table 1 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161> reports sample characteristics. The characteristics of respondents P-MIC sample were compared with those of the general Australian population in a previous manuscript,²⁸ finding that, although there were small differences, characteristics were largely comparable. The mean age was 14.88 years (SD 1.87), 42.47% respondents ($n = 251$) had SHCNs, and 36.11% respondents had clinically important MHCs ($n = 208$).

Acceptability and Feasibility

Over 70% of the participants reported that both instruments were easy ("somewhat" or "very" easy) to complete (see Appendix Table 2 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161>). A similar percentage of respondents reported that Y-5L and 5L were difficult ("somewhat" or "very" difficult) to complete (6.77% vs 6.26%). There was no statistically significant difference between participant-reported difficulty completing the Y-5L and 5L (Chi2 1.71, *P* value .789).

For the SHCN, MHC, and age subgroups (12–13, 14–16, and 17–18), 7.97%, 10.09%, 6.79%, 5.74%, and 6.77% of adolescents, respectively, reported difficulty completing the 5L, compared with 9.57%, 13.94%, 6.17%, 6.42%, and 8.27% for the Y-5L, respectively. These differences were statistically significant.

The time to complete both instruments has a positively skewed distribution for all participants and subgroups (see Appendix Table 3 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161>). On average, respondents spent 30.63 seconds completing the 5L and 41.49 seconds the Y-5L. There were no statistically significant differences in completion time by SHCN, MHC, or age group.

Response Distribution and Floor and Ceiling Effects

The response distributions for the full sample and subgroups are displayed in Appendix Table 4 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161>. On the 5L, 10.32%, 8.29%, 22.84%, 41.62%, and 52.62% reported some problems (level 2 or above) on MO, SC, UA, PD, and AD, respectively, whereas 13.03%, 9.31%, 29.92%, 45.18%, and 62.61% reported some problems on the corresponding dimensions of the Y-5L. The differences in response distribution in each dimension between the 2 instruments were statistically significant in the full sample and each subgroup. The differences between the 2 instruments were larger in the 12-to-13-year-old age group. Among adolescents with SHCN or MHC, higher proportions of problems (and more severe problems) were reported compared with those in the full sample. Over 90% of respondents who reported level 5 on AD/WSU have been identified as having a clinically important MHC (see Appendix Table 5 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161>). In addition, we found large differences between the 2 instruments in reporting severe problems (ie, level 4 or 5) on UA, PD in the SHCN group, and on PD and AD/WSU in the MHC group (see Appendix Table 4 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161>). For example, 4.78% of respondents with SHCN reported level 4 or 5 on UA in the 5L, compared with 10.75% on the Y-5L. Among adolescents with MHC, 4.81% and 10.09% of respondents reported “severe” or “extreme” pain on 5L and Y-5L, respectively, and 15.86% of respondents reported “severely” or “extremely” AD on the 5L compared with 21.63% WSU on the Y-5L.

Figure 3 presents Sankey plots showing how response distribution by dimension differed between Y-5L and 5L for the full sample. Overall, most variability is in the UA, PD, and AD/WSU dimensions. Among adolescents with SHCN or MHC, there is more variability in severity of AD/WSU dimension compared with the full sample (see Appendix Fig. 1 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161>). There is also more dispersion across levels for this dimension compared with the UA and PD.

At the instrument level, a ceiling effect was evident for the 5L (19.12%) only in the SHCN subgroup. Although there was no ceiling effect for Y-5L, at 14.74% the SHCN subgroup approached the

cutoff. There are no floor effects for either instrument (see Appendix Table 4 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161>). Appendix Figure 2 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161> plots the distribution of LSSs for each instrument for the full sample and by subgroup, which showed a positively skewed frequency distribution.

Convergent Validity

There is a strong positive correlation between the corresponding dimensions of the Y-5L and 5L for all respondents (Spearman ρ , range: 0.63–0.74, all $P < .001$) (see Appendix Table 6 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161>). Subgroup analyses reflected similarly strong positive correlations between expected dimensions between the 2 instruments (see Appendix Table 7 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161>). The correlation coefficients were larger in the 17-to-18-year-old subgroup between expected dimension pairs compared with younger age groups, and we observed strong correlations between multiple dimensions. In particular, UA of 5L was strongly correlated to all dimensions of Y-5L, except for PD, which was moderately correlated ($\rho = 0.51$ – 0.63 vs $\rho = 0.44$).

Test-Retest Reliability

As shown in Table 1, both the 5L and Y-5L dimensions demonstrated fair to moderate agreement between initial and follow-up tests (Table 1). There was, at most, moderate agreement, with SC (weighted kappa = 0.5045) and PD (weighted kappa = 0.4761) on the 5L and MO, WSU (both with weighted kappa = 0.4107), and UA (weighted kappa = 0.4310) on the Y-5L showing the highest agreement.

The 2-way mixed-effects model, ICC, shows good to excellent agreement for test-retest reliability for the 5L LSS among all participants (ICC = 0.641) and among those with SHCN (ICC = 0.757), with only fair agreement for the Y-5L LSS (ICC = 0.454, ICC = 0.456, respectively). We found stronger test-retest reliability for both instruments in the 12-to-13-year-old subgroup (5L ICC = 0.857, Y-5L ICC = 0.615) compared with older adolescents (5L ICC = 0.541, Y-5L ICC = 0.478 for 14–16-year-olds; 5L ICC = 0.593, Y-5L = 0.399 for 17–18-year-olds).

Known-Group Validity

Table 2 reports the known-group validity of both instruments. LSS for both were able to detect a significant difference between those with and without SHCN, and Y-5L had a stronger effect size (Cohen's $d = -0.78$ for the 5L and -0.86 for Y-5L). The Y-5L and the 5L demonstrated a moderate effect size in discriminating

Figure 3. Sankey plots, demonstrating the descriptive relationship between self-reported health of adolescents on the EQ-5D-Y-5L(left) versus EQ-5D-5L (right) plots, all participants.

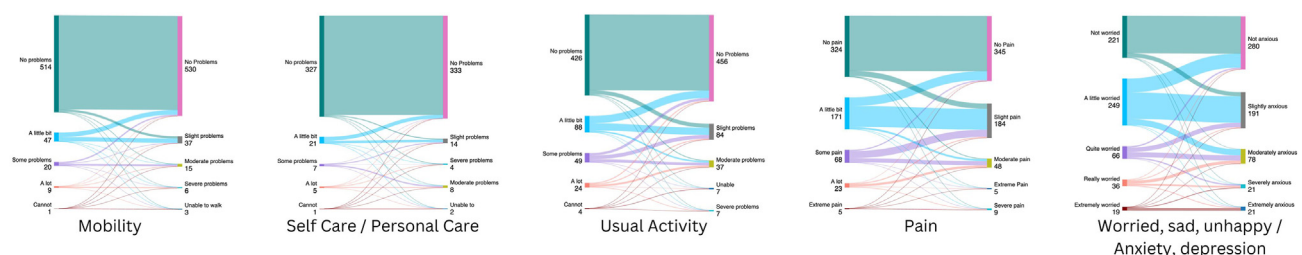


Table 1. Test-retest results for EQ-5D-5L and EQ-5D-Y-5L, compared by health and age subgroups among those had no change in health between baseline and follow-up.

HRQoL dimensions and level sum score	Full (N = 486)		SHCN (N = 207) [§]		MHC (N = 177) [§]		Age subgroup					
							12-13-year-olds (N = 137)		14-16-year-olds (N = 234)		17-18-year-olds (N = 115)	
	Agreement	Weighted kappa*	Agreement	Weighted kappa*	Agreement	Weighted kappa*	Agreement	Weighted kappa*	Agreement	Weighted kappa*	Agreement	Weighted kappa*
EQ-5D-5L												
MO	87.27%	0.3668	78.79%	0.4978 [†]	78.38%	0.4667 [†]	93.94%	0.4762 [†]	85.71%	0.3864	82.14%	0.2308
SC	92.73%	0.5045 [†]	87.88%	0.6196 [†]	89.19%	0.6263 [†]	100.00%	1 [†]	87.76%	0.1009	92.86%	0.7241
UA	75.45%	0.2984	63.64%	0.3582	56.76%	0.2487	81.82%	0.3356	73.47%	0.3378	71.43%	0.1734
PD	74.55%	0.4761 [†]	63.64%	0.3982	56.76%	0.3092	72.73%	0.3901	75.51%	0.5273 [†]	75.00%	0.4571
AD	64.55%	0.3682	51.52%	0.3187	37.84%	0.1369	72.73%	0.449 [†]	61.22%	0.3052	60.71%	0.3913
	ICC [‡]	ICC range	ICC [‡]	ICC range	ICC [‡]	ICC range	ICC [‡]	ICC range	ICC [‡]	ICC range	ICC [‡]	ICC range
LSS	0.641**	[0.515 0.739]	0.757***	[0.56 0.872]	0.592*	[0.333 0.768]	0.857***	[0.729 0.927]	0.541*	[0.309 0.713]	0.593*	[0.285 0.789]
EQ-5D-Y-5L												
	Agreement	Weighted kappa*	Agreement	Weighted kappa*	Agreement	Weighted kappa*	Agreement	Weighted kappa*	Agreement	Weighted kappa*	Agreement	Weighted kappa*
MO	86.92%	0.4107 [†]	80.65%	0.5452 [†]	77.78%	0.5034 [†]	91.18%	0.4688 [†]	82.22%	0.3023	82.14%	0.2308
SC	87.85%	0.2345	74.19%	0.1788	77.78%	0.191	91.18%	−0.0303	88.89%	0.245	92.86%	0.7241
UA	78.50%	0.4310 [†]	58.06%	0.2867	66.67%	0.4454 [†]	82.35%	0.4237 [†]	77.78%	0.4649 [†]	71.43%	0.1734
PD	68.22%	0.3894	51.61%	0.2488	55.56%	0.301	73.53%	0.4632 [†]	68.89%	0.435 [†]	75.00%	0.4571
WSU	86.92%	0.4107 [†]	45.16%	0.1804	30.56%	−0.0477	61.76%	0.305	62.22%	0.3855	60.71%	0.3913
	ICC [‡]	ICC range	ICC [‡]	ICC range	ICC [‡]	ICC range	ICC [‡]	ICC range	ICC [‡]	ICC range	ICC [‡]	ICC range
LSS	0.454*	[0.290 0.592]	0.456*	[0.134 0.692]	0.314	[−0.005 0.577]	0.615**	[0.346 0.790]	0.478*	[0.222 0.673]	0.399	[0.030 0.670]

AD indicates anxiety/depression; LSS, level sum scores; MHC, mental health concerns; MO, mobility; PD, pain or discomfort; SC, self-care; SHCN, special health care needs; UA, usual activities; WSU, worried, sad, unhappy.

*A weighted kappa of 0.2 indicates poor agreement; 0.21-0.40, fair agreement; 0.41-0.6, moderate agreement; 0.61-0.80, substantial agreement; >0.81, almost perfect agreement.²⁵

[†]Indicates moderate agreement.

[‡]ICC values below 0.4 indicate poor agreement; between 0.40 and 0.59 indicate fair agreement and are denoted by *; between 0.60 and 0.74 indicate good agreement and are denoted by **; and between 0.75 and 1 indicate excellent agreement and are denoted by ***.²⁶

[§]The SHCN and MHC groups are not mutually exclusive. N = 107 were members of both groups. Analysis of the data in terms of 3 groups (SHCN only, MHC only, and those with both SHCN and MHC), did not change conclusions about the performance of Y-5L versus 5L.

SHCN among adolescents aged 14 to 16 years (Cohen's $d = -0.73$ and -0.77 , respectively), but a strong effect size in discriminating SHCN among the younger and older adolescent subgroups. Both Y-5L and 5L differentiated adolescents' HRQoL by MHC status, with Y-5L demonstrating stronger effect size across all age groups.

Summary of Results

Results of all analyses are summarized in Table 3. Both the Y-5L and the 5L showed good psychometric performance, with Y-5L meeting 3 of 4 prespecified cutoffs in the full sample and the 5L meeting 2 of 4. The Y-5L slightly outperformed the 5L on ceiling effects and known-group validity; the 5L slightly outperformed the Y-5L on test-retest reliability.

Discussion

Based on current advice in children 12 to 15 years of age, either the Y-5L or the 5L can be used.¹⁰ Although the 5L is recommended for children aged 16 to 18, guidance indicates that either could be used, depending on the study requirements.¹⁰ Therefore, in the collection and use of these data for cost-effectiveness models, the transition point between child and adult HRQoL could in principle occur anywhere between 12 and 18 years of age. Researchers face a choice about which instrument to use within this range and how

to handle the transition between them in terms of measurement and valuation. There is some evidence on the measurement properties of the 3-level versions of these instruments (ie, the adult 3L and Y-3L) in children/adolescents, eg, in 15-to-17-year-old adolescents in China, where respondents reported more problems on the Y-3L compared with the 3L, as well as lower ceiling effects, which the authors attributed to differences in dimension wording.²⁹ Similarly, a comparison of the 13-to-17-year-old adolescents in South Africa³⁰ found more problems reported on the Y-3L compared with the 3L. A study of 13-to-18-year-olds in Ethiopia and South Africa comparing both Y-3L with 3L and Y-5L with 5L found that adolescents reported more problems on the youth instruments, but this difference was less marked for the 5-level versions.³¹ However, the authors noted limitations to the generalizability of their findings because of local adaptations that had been made to the youth instruments and recommended that future work is needed to investigate the differences in the Y-5L and 5L in English. Our article has addressed this need and, to our knowledge, provides the first evidence on the comparative psychometric performance of the English-language versions of the Y-5L and 5L in a population of 12 to 18 year olds.

Overall, our results suggest that the Y-5L has somewhat better psychometric performance than the 5L in the ages 12 to 18 years. However, the extent to which one instrument is considered better than the other therefore depends on a judgement about the

Table 2. Known-group validity of the EQ-5D-5L and EQ-5D-Y-5L.

Known groups	N	EQ-5D-5L			EQ-5D-Y-5L		
		Mean	Difference	ES	Mean	Difference	ES
		(SD)			(SD)		
Full sample (N = 591)*							
SHCN							
No	340	6.17 (1.90)	−2.07	−0.78	6.46 (1.92)	−2.33	−0.86
Yes	251	8.24 (3.39)			8.79 (3.51)		
MHC							
No	368	6.13 (1.72)	−2.37	−0.93	6.31 (1.79)	−2.89	−1.16
Yes	208	8.50 (3.46)			9.20 (3.41)		
12-13 years (N = 162)							
SHCN							
No	105	6.05 (1.7)	−2.29	−0.82	6.40 (1.80)	−2.16	−0.83
Yes	57	8.33 (4.07)			8.56 (3.66)		
MHC							
No	106	6.00 (1.84)	−2.50	−0.91	6.15 (1.47)	−2.98	−1.22
Yes	53	8.50 (4.02)			9.13 (3.66)		
14-16 years (N = 296)							
SHCN							
No	164	6.28 (2.00)	−1.82	−0.73	6.70 (2.16)	−2.15	−0.77
Yes	132	8.10 (2.97)			8.84 (3.44)		
MHC							
No	183	6.29 (1.90)	−1.97	−0.82	6.55 (2.02)	−2.71	−1.06
Yes	107	8.26 (3.04)			9.26 (3.27)		
17-18 years (N = 133)							
SHCN							
No	71	6.10 (1.92)	−2.35	−0.83	6.01 (1.38)	−2.87	−1.09
Yes	62	8.45 (3.59)			8.89 (3.56)		
MHC							
No	79	5.94 (1.60)	−3.06	−1.18	6.01 (1.14)	−3.12	−1.33
Yes	48	9.00 (3.68)			9.13 (3.50)		

Note. Because of the skewed distribution of LSS, the mean difference in LSS between groups is tested using nonparametric Mann-Whitney U test. $P < .001$ for all tests. Effect sizes of 0.2-0.49 are considered small, 0.5-0.79 moderate, and ≥ 0.8 large.²⁷

ES indicates effect size; MHC, adolescents with mental health concern, identified by Strengths and Difficulties Questionnaire²²; SHCN, adolescents with special health care needs, identified by Child Special Health Care Screener.²⁰

*The SHCN and MHC groups are not mutually exclusive. $N = 126$ were members of both groups. Analysis of the data in terms of 3 groups (SHCN only, MHC only, and those with both SHCN and MHC) did not change conclusions about the performance of Y-5L versus 5L.

relative importance of these different aspects of psychometric performance. Both were easy and quick for respondents in this age group to complete. Acceptability, feasibility, and convergent validity were good across both instruments. Ceiling effects were not observed for the Y-5L among adolescents with MHC or SHCN but were observed for the 5L in the latter. The Y-5L better differentiated between adolescents with and without SHCN and between groups defined by MHC than the 5L. The 5L showed better test-retest reliability in adolescents with SHCN and MHC. Spearman's correlation coefficients indicated strong convergent validity between each of the 5 related dimensions on the Y-5L and 5L. This is as expected, given that the 2 cover the same 5 dimensions, albeit using more appropriate, child-friendly wording for the Y-5L.^{2,10}

However, these child-friendly adaptations, aimed at improving the relevance of health-state descriptions to children via both the dimension and level descriptors (Fig. 1) and to improve their ability to self-report their health have the potential to subtly alter the underlying construct that is being measured. Our results suggest that these differences in wording lead to differences in the HRQoL being measured.

For example, one of the lowest correlations we observed between the 5L and Y-5L dimensions is for UA. The description of UA in the 5L is "work, study, housework, family, or leisure activities," whereas for the Y-5L it is "going to school, hobbies, sports, playing, doing things with family or friends." Younger adolescents may struggle to comprehend the concept of work and the definition of leisure in the 5L. Even if provided with a definition of leisure, they may view it as separate from the concept of play or doing things with friends.

The differences between the Y-5L and the 5L appear most marked in the case of the WSU dimension versus "anxious/depressed." In the overall sample, the proportion of adolescents reporting "no problems" on this dimension is 20% lower on the Y-5L than on the 5L (see Appendix Table 4 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.04.2161>), and the proportion reporting mild problems (level 2) is 30% higher on the Y-5L than on the 5L. These differences reinforce findings from previous studies comparing adult and youth EQ-5D instruments.^{29,31} It is unclear to what extent these differences are a product of the differences in dimension descriptors, level labeling,

Table 3. Summary of psychometric performance of the EQ-5D-5L and the EQ-5D-Y-5L for the combined sample and by subgroup.

Psychometric properties	EQ-5D-5L	EQ-5D-Y-5L
Acceptability/feasibility		
EQ-5D-5L vs EQ-5D-Y-5L Chi-square (<i>P</i> value)		
All respondents	✓764.62 (<.0001)	
SHCN*	✓357.94 (<.0001)	
SHCN, nil MHC	✓168.45 (<.0001)	
MHC*	✓295.82 (<.0001)	
MHC, nil SHCN	✓77.69 (<.0001)	
SHCN and MHC	✓204.56 (<.0001)	
12-13-year-olds	✓216.39 (<.0001)	
14-16-year-olds	✓357.15 (<.0001)	
17-18-year-olds	✓254.86 (<.0001)	
Ceiling effects [†]		
All respondents		
SHCN	✗19.12%	✓14.74%
MHC	✓13.46%	✓14.74%
SHCN and MHC	✓8.73%	✓6.35%
12-13-year-olds		
12-13-year-olds, SHCN	✗22.81%	✗15.79%
12-13-year-olds, MHC	✓13.21%	✓7.55%
14-16-year-olds		
14-16-year-olds, SHCN	✗18.94%	✗15.91%
14-16-year-olds, MHC	✗15.89%	✓10.28%
17-18-year-olds		
17-18-year-olds, SHCN	✗16.13%	✓11.29%
17-18-year-olds, MHC	✓8.33%	✓10.42%
Convergent validity		
All respondents	✓	✓
SHCN	✓	✓
MHC	✓	✓
SHCN and MHC	✓	✓
12-13-year-olds	✓	✓
14-16-year-olds	✓	✓
17-18-year-olds	✓	✓
Known-group validity		
All respondents, by SHCN	✗	✓
All respondents, by MHC	✓	✓
Among 12-13-year-olds, by SHCN	✓	✓
Among 12-13-year-olds, by MHC	✓	✓
Among 14-16-year-olds, by SHCN	✗	✗
Among 14-16-year-olds, by MHC	✓	✓
Among 17-18-year-olds, by SHCN	✓	✓
Among 17-18-year-olds, by MHC	✓	✓
Test-retest reliability		
All respondents	✓	✗

Continued in the next column

Table 3. Continued

Psychometric properties	EQ-5D-5L	EQ-5D-Y-5L
SHCN	✓	✗
MHC	✗	✗
SHCN and MHC	✓	✗
12-13-year-olds	✓	✓
14-16-year-olds	✗	✗
17-18-year-olds	✗	✗

Note. Data presented in this table compare specific metrics of performance where available. All results for all subgroups are provided in the [Supplemental Materials \(Tables S1-S7\)](https://dx.doi.org/10.1016/j.jval.2025.04.2161) found at <https://dx.doi.org/10.1016/j.jval.2025.04.2161>. MHC indicates mental health concerns; SHCN, special health care needs.

Key for interpreting each cell in the table:

✓ = Good performance ✗ = Poor performance

Acceptability/feasibility: Differences in difficulty were assessed using Pearson's chi-square test.

Ceiling effects: Percentage of respondents reporting the best possible health state (level sum score of 5). A ceiling effect is present if >15% of participants scored in the best possible health state.⁶

Test-retest reliability: Denoted with a green tick if the ICC value was good or excellent, according to the thresholds recommended by Cicchetti (1994)²⁶; denoted with red x if the ICC value was poor or fair.

Known groups: Denoted with a green tick if mean difference with a value of <0.05 and large effect size (Cohen's *d* effect size ≥ 0.8).²⁷

Convergent validity: Indicates that dimensions on the EQ-5D-5L were strongly correlated to expected dimensions on the EQ-5D-Y-5L (Spearman's correlation ≥ 0.5).

*The SHCN and MHC groups are not mutually exclusive. *N* = 126 were members of both groups. Analysis of the data in terms of 3 groups (SHCN only, MHC only, and those with both SHCN and MHC), did not change conclusions about the performance of Y-5L versus 5L.

[†]There were no floor effects observed for either measurement in this study, for the combined group, or in any subgroup. A floor effect is present if >15% of participants scored in the worst possible health state (level sum score of 25).

or both. Interestingly, the proportion of adolescent respondents reporting level-5 problems (labeled “extreme problems” on both the Y-5L and 5L) is very similar on both instruments. Overall, however, there are important differences in the measurement of HRQoL and the suggested prevalence of mental health issues. It suggests that WSU means something different to respondents self-reporting their health than “anxious or depressed.” The former may be seen as representing “less severe” kinds of problems and therefore picks up a greater prevalence of issues. A related possibility is that AD may be linked to HRQoL, whereas WSU may be picking up a wider range of (nonhealth-related) day-to-day influences on mental state.³²

The differences we have identified between the prevalence and distribution of HRQoL problems captured by the 2 instruments could potentially lead to different conclusions being drawn about the impact of illness and treatment on HRQoL, depending on which instrument is used. In situations in which these differences exist, a transition from using Y-5L to the 5L (eg, in a longitudinal study or in an economic model) could lead to a change in self-reported HRQoL, even if the child's underlying health state has remained unchanged.

The implications for the assessment of changes in HRQoL and incremental QALY gains in the assessment of relative effectiveness and cost-effectiveness of treatments are unknown. Further research is needed to identify the extent to which the measurement differences we identify “wash out” when the focus is on change between baseline and follow-up assessments. Furthermore, the assessment of HRQoL in economic models is a product of both the measurement properties of these instruments and the preference weights applied to summarize the

health states described by them. Depending on the characteristics of values for the Y-5L that emerge in future value set studies and how these compare with value sets for the 5L (eg, in terms of scale length and the weights attached to dimensions), the application of preference weights could either exacerbate or moderate the differences in HRQoL measurement that we have identified. The combination of measurement and valuation differences have the potential to create an artefactual cliff edge¹¹ in the assessment of HRQoL across ages, which is relevant both to studies involving a transition between age-specific instruments and, more widely, to comparisons of estimates of QALY gains and cost-effectiveness based on age-specific HRQoL instruments. A research program to explore this occurrence, once Y-5L values are available, is required.

One limitation of our article is that we were unable to assess the responsiveness of the 2 instruments because of the limited number of adolescents reporting improved or worsened health between baseline and follow-up. Further evidence on responsiveness is required. Because of data being collected online, a further limitation is that we cannot be sure if the adolescents completed the HRQoL instruments with the assistance of their caregiver or not. Although this likely reflects the real-world nature of collecting HRQoL data, further research may be needed to assess if there are differences between adolescent responses in cases in which they had assistance from a caregiver versus no assistance. Users and analysts need to be aware of the potential for discontinuities in the measurement of HRQoL across age-specific instruments. Our article has identified the potential for discontinuities between age-specific instruments even when they have closely related descriptive systems; the challenges are likely to be greater when comparing unrelated age-specific instruments of HRQoL.

The implications for the estimation of QALYs will depend both on differences in the measurement of HRQoL, such as those identified in this article, and on differences in the age-specific preference weights (values) applied to summarize HRQoL.¹¹ Another limitation of our article is that we have restricted our analysis to the way HRQoL is self-reported by adolescents on each instrument. The characteristics of the corresponding value sets for the Y-5L and 5L will introduce an exogenous source of variation to analysis and comparability of these data.³³ Value sets for the Y-3L have been shown to have markedly different characteristics than value sets for the adult 5L.³⁴ Value sets for the Y-5L are not yet available; therefore, the overall implications for preference-weighted HRQoL remains a question to be explored in future.

Conclusions

Results support both the Y-5L and adult 5L as instruments for measuring HRQoL in 12 to 18 year olds, with Y-5L having some psychometric advantages. The psychometric performance of each differed somewhat both for the sample overall (eg, a lower prevalence of “no problems” on the Y-5L than on the 5L) and between subgroups. Choosing between these instruments for this age group should therefore be guided by the specific application. Despite both instruments sharing broadly the same 5-dimension, 5-level structure, the HRQoL data produced by each present a somewhat different picture of the sample's HRQoL. This is linked to the adaptations made to the Y-5L to make it child friendly. Users should be aware that transitions between age-specific instruments of HRQoL have the potential to generate discontinuities in measurement of HRQoL and that comparisons of HRQoL

between age-specific instruments should be approached thoughtfully and with care.

Author Disclosures

Author disclosure forms can be accessed below in the [Supplemental Material](#) section.

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Author Affiliations: Child Health Economics Unit, Melbourne School of Population and Global Health, University of Melbourne, Melbourne, VIC, Australia (Reyes, Pan, Jones, Dalziel, Devlin); Health Services and Economics, Murdoch Children's Research Institute, Melbourne, VIC, Australia (Jones, Dalziel).

Correspondence: Nicole Reyes, MPH, 2113/50 Albert Road, South Melbourne, VIC, 3205, Australia. Email: nicole.a.reyes@hotmail.com

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